

Microsoft Fabric Warehouse

Database Mirroring

DBA Technical Reference | Volume 8 of 9

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1. What is Fabric Database Mirroring?

Fabric Database Mirroring is a near-real-time data replication feature that continuously synchronizes data from external databases into Fabric OneLake, making it available as Delta Lake tables that can be queried by the Warehouse and other Fabric experiences.

Key benefits of mirroring:

- Near-zero latency data availability — changes in source appear in Fabric within seconds to minutes
- No ETL pipeline code required — mirroring is configuration-driven
- Data is stored as open Delta Lake format — queryable by all Fabric experiences
- No impact on source database performance — mirroring uses CDC log reading
- Eliminates need for nightly data extracts from operational systems

INFO: Mirrored data lands in a Fabric Lakehouse (as Delta tables) and is queryable via the Warehouse's cross-database T-SQL capabilities. Mirroring does not write directly into a Warehouse.

2. Supported Source Databases

Source Database	Support Status	Prerequisites
Azure SQL Database	Generally Available	Enable CDC on source database
Azure SQL Managed Instance	Generally Available	Enable CDC; outbound connectivity
SQL Server 2016+	Preview	SQL Server with CDC + network access to Fabric
Azure Cosmos DB	Generally Available	Enable Change Feed on containers
Azure Database for PostgreSQL	Preview	Enable logical replication
Snowflake	Preview	Snowflake account with appropriate permissions
Azure Databricks	Preview	Delta Sharing enabled on source catalog

3. How Mirroring Works (CDC-Based Replication)

Fabric mirroring uses Change Data Capture (CDC) to track row-level changes (INSERT, UPDATE, DELETE) in the source database transaction log and replicate them to OneLake as Delta Lake operations.

Phase	Description
Initial snapshot	Full table scan of source — all existing rows copied to OneLake as Delta tables
CDC initialization	CDC is enabled on source tables; change tracking begins from snapshot LSN
Continuous replication	INSERT/UPDATE/DELETE changes read from CDC log and applied to Delta tables
Delta commit	Changes committed as new Delta Lake log entries — immediately queryable
Compaction	Background compaction periodically optimizes Delta table file layout

4. Setting Up Mirroring from Azure SQL Database

Step 1: Enable CDC on the Source Azure SQL Database

>> Enable CDC on Azure SQL Database

```
-- Run on the SOURCE Azure SQL Database (not Fabric)

-- Enable CDC at database level
EXEC sys.sp_cdc_enable_db;

-- Enable CDC on each table to be mirrored
EXEC sys.sp_cdc_enable_table
@source_schema = 'dbo',
@source_name = 'Orders',
@role_name = NULL,
@supports_net_changes = 1;

-- Verify CDC is enabled
SELECT name, is_cdc_enabled
FROM sys.databases
WHERE name = DB_NAME();
```

Step 2: Create the Mirror in Fabric Portal

Step 1: In Fabric workspace, click + New > Mirror database

Step 2: Select Azure SQL Database as the source type

Step 3: Enter the Azure SQL Server name, database name, and credentials

Step 4: Select which tables to include in the mirror

Step 5: Click Create mirror — initial snapshot begins immediately

Step 6: Monitor snapshot progress in the Mirror properties panel

5. Setting Up Mirroring from SQL Server

Step 1: Configure SQL Server for CDC

>> Enable CDC on SQL Server

```
-- Run on the ON-PREMISES or Azure VM SQL Server

-- Enable CDC at database level
USE YourDatabase;
EXEC sys.sp_cdc_enable_db;

-- Enable CDC on tables
EXEC sys.sp_cdc_enable_table
@source_schema = 'dbo',
@source_name = 'Customers',
@role_name = NULL;

-- Verify SQL Server Agent is running (required for CDC)
SELECT name, status_desc
FROM sys.dm_server_services
WHERE servicename LIKE 'SQL Server Agent%';
```

Step 2: Configure Network Connectivity

- Install the Fabric On-premises Data Gateway on a server with access to SQL Server
- Ensure the gateway server can reach both the SQL Server instance and Fabric service endpoints
- Open required firewall ports for the Data Gateway outbound connection
- Register the gateway in Fabric Admin portal and assign it to the workspace

6. Monitoring Mirroring Status

Monitoring Method	What to Check
Fabric portal > Mirror item	Overall mirror status, last sync time, table-level status
Mirror replication monitoring view	Rows replicated, lag time, error details
Fabric Capacity Metrics App	CU consumption from mirroring workload
Source database CDC tables	sys.dm_cdc_log_scan_sessions on source

>> Query Mirroring Status (Fabric Side)

```
-- Query the mirrored Lakehouse to verify data freshness
SELECT
MAX(LastModifiedDate) AS latest_record,
COUNT(*) AS total_rows,
GETUTCDATE() AS current_utc
FROM LakehouseName.dbo.MirroredTableName;

-- Compare with expected latency threshold
-- If latest_record is > 5 minutes behind GETUTCDATE() investigate lag
```

7. Mirroring Latency & Performance

Latency Factor	Typical Range	Optimization
Initial snapshot	Minutes to hours	Run during off-peak hours for large tables
Ongoing CDC replication	Seconds to minutes	Reduce source transaction volume; optimize network
Delta compaction lag	Minutes	Normal — compaction runs asynchronously
Network latency	Varies	Use same Azure region for source and Fabric where possible
Large transactions	Higher lag	Break large batch operations into smaller transactions

8. Mirroring Limitations

Limitation	Details
Tables without primary keys	Tables must have a primary key to be mirrored
DDL changes	Schema changes on source (ALTER TABLE) may require mirror reset
TRUNCATE TABLE	TRUNCATE is not captured by CDC — use DELETE instead on mirrored tables
BLOB/LOB columns	Large object columns (VARBINARY(MAX), TEXT) may not be mirrored
DR replication	Mirrored databases are NOT included in BCDR OneLake replication
Write back	Mirrored data in Lakehouse is read-only — cannot write back to source via Fabric

9. Mirroring vs Other Integration Patterns

Pattern	Latency	Complexity	Use Case
Fabric Mirroring	Seconds - minutes	Low (config only)	Near-real-time operational reporting
Data Factory COPY	Minutes - hours	Medium (pipeline code)	Batch ETL, transformations needed
Dataflow Gen2	Minutes - hours	Medium (Power Query)	Smaller datasets, business user ETL
Spark Notebooks	Variable	High (code)	Complex transformations, ML feature engineering
COPY INTO T-SQL	Minutes	Low (SQL)	File-based loads from storage

10. Mirroring Best Practices

- Ensure all tables to be mirrored have a primary key before enabling CDC
- Run the initial snapshot during off-peak hours to minimize source database impact
- Monitor CDC log growth on source — large unreplicated CDC logs can impact source performance
- Use the same Azure region for source database and Fabric capacity to minimize replication latency
- Test mirror failover recovery as part of your DR drill — mirrors must be recreated post-regional-failover
- Do not use TRUNCATE on mirrored source tables — it is not captured by CDC; use DELETE instead

- Set up Azure Monitor alerts on the source database CDC agent job failures